

Roll No. _____

B.Tech IV sem (Main/Back) Exam April-May 2026**Artificial Intelligence & Data Science****4AID3-04 Microprocessor & Interfaces****Paper Code : 4E1304****CS, AID, CAI, CCS, CDS, CIT, CSR****Time: 3 Hours****Maximum Marks: 70**

Attempt all ten questions from Part A, five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

1. _____

2. _____

Part A (Answer should be given up to 25 words only)**All questions are compulsory**

T-483

- Q.1 What is a microprocessor based computer?
 Q.2 Explain de-multiplexing of buses.
 Q.3 What is meant by liquid crystal display?
 Q.4 Why do you need 8255 chip?
 Q.5 What are direct addressing modes?
 Q.6 What is an instruction cycle?
 Q.7 Is there a relation between T state, machine cycle & frequency?
 Q.8 What are serial/parallel interfaces?
 Q.9 Define macro in microprocessor 8085.
 Q.10 What do you mean by machine language?

10 x 2 = 20**Part B (Analytical/Problem solving questions)****Attempt any Five questions**

- Q.11 State the difference between Trap and other interrupts.
 Q.12 State the difference between Centronics and IEEE488 parallel interfaces.
 Q.13 Explain the classification of instructions in general.
 Q.14 Draw the Flag register format and explain the individual flags.
 Q.15 Explain memory interfacing in microprocessor 8085.
 Q.16 Explain RIM format.
 Q.17 Write a program to subtract two 8-bit numbers.

5x 4 = 20

Part C (Descriptive/Analytical/Problem Solving/Design question)

Attempt any three questions

- Q.18 Write a program to add seven 8-bit numbers located at C020 onwards. Place the result at location C200 onwards.
- Q.19 Describe the working of 8279 chip.
- Q.20 Explain the different modes of working of 8259 chip.
- Q.21 What is the use of USART 8251? Explain its working. <https://www.pyqonline.com>
- Q.22 Write a program to place a list of 6 data from location C400 onwards to C700 onwards.

3 x 10= 30